Deep vein thrombosis (DVT) is the formation of a thrombus (blood clot) in the deep veins, most often of the lower limb (calf, popliteal, femoral, iliac veins). Its clinical importance is the risk of **pulmonary embolism (PE)** when the clot detaches.

## 1. Pathogenesis: Virchow's triad

Thrombosis results from one or more of three factors:

- **Venous stasis:** immobility (bed rest, long flights, paralysis), heart failure, obesity, pregnancy, varicose veins. Stasis lets activated clotting factors accumulate and brings platelets into contact with the endothelium. Clots often start in valve pockets, where flow is slowest.
- **Endothelial injury:** surgery, trauma, central lines, prior DVT, inflammation. Exposed subendothelial collagen and tissue factor activate coagulation.
- **Hypercoagulability:**
    - *Inherited:* factor V Leiden (most common), prothrombin G20210A mutation, deficiency of protein C, protein S, or antithrombin.
    - *Acquired:* malignancy, pregnancy and postpartum, oral contraceptives and estrogen therapy, nephrotic syndrome, antiphospholipid syndrome, heparin-induced thrombocytopenia, major surgery.

## 2. Thrombus formation

Venous thrombi are **red (stasis) thrombi**, rich in fibrin and red cells with fewer platelets, unlike arterial thrombi, which are platelet-rich white thrombi.

- Coagulation is triggered (tissue factor pathway and contact activation), generating thrombin.
- Thrombin converts fibrinogen to fibrin, forming a mesh that traps red cells.
- The thrombus propagates proximally along the vein, often in the direction of flow. Poorly attached propagating tails are the part that embolizes.

## 3. Fate of the thrombus

- Propagation up the vein
- **Embolization** to the lungs (PE)
- Dissolution by fibrinolysis
- Organization and recanalization, leaving scarred, damaged valves

- Rarely, infection

## 4. Clinical features

Many cases are asymptomatic. When present:

- Unilateral leg swelling, pain or tenderness (often calf), warmth, erythema, and pitting edema
- Dilated superficial collateral veins
- Calf circumference difference of more than 3 cm compared with the other leg is a useful sign
- Homan's sign is unreliable and not recommended

## 5. Diagnosis

- **Wells score** to estimate pre-test probability
- **D-dimer:** high sensitivity, low specificity, so a negative result rules out DVT only when probability is low
- **Compression duplex ultrasound:** the main imaging test, where a non-compressible vein indicates thrombus
- Venography is rarely needed

## 6. Management

- **Anticoagulation** is the cornerstone: low-molecular-weight heparin or fondaparinux, then or alongside direct oral anticoagulants (rivaroxaban, apixaban) or warfarin. Treatment lasts at least 3 months, longer if unprovoked or ongoing risk.
- **IVC filter** only if anticoagulation is contraindicated.
- **Catheter-directed thrombolysis** in selected cases such as extensive iliofemoral DVT.
- **Compression stockings** for symptom relief; their role in preventing post-thrombotic syndrome is now debated.

## 7. Prevention

Early mobilization, hydration, mechanical measures (intermittent pneumatic compression, stockings), and pharmacological prophylaxis (LMWH or DOACs) in high-risk hospitalized or surgical patients.

## 8. Complications

- **Pulmonary embolism**
- **Post-thrombotic syndrome:** chronic pain, swelling, and skin changes from valve damage and venous hypertension
- Recurrent DVT
- Phlegmasia cerulea dolens (severe limb-threatening obstruction), a rare complication

**Key point:** Since DVT often arises in the calf and propagates proximally, proximal DVT (popliteal and above) carries the highest PE risk and is treated most aggressively.